Design Approach

March 29, 2026 2:41 PM

Define constraints:

- Total power limit $\leq 1\,mV$ at $V_{DD} = 3.3V$
- Total current: $1mV/3.3V = 303\mu A$
- Max branch current $\leq 200\mu A$ / branch
  - To stay safe, allocate $80-90\mu A$ for each stage.

Stage 1 Design:

- Lets take this to be common source
  - Provides voltage gain and high input resistance required. ($\geq 100k\Omega$)
- since gate draws no DC current,
  - R_{in} is determined by biasing resistors R_1, R_2.
  - To meet requirement, $R_1 \| R_2 \geq 100k\Omega$

- Lets choose $V_{ov} = 0.2V$
  - Within range $(0.2-0.6)$
  - low V_{ov} = large g_m → Easier to hit 60dB gain w/o exceeding power.

- Assume NMOS
- Given NMOS parameters:
  - $V_{tn} = 0.63V$
  - $\mu_n C_{ox} = 100\mu A/V^2$
  - Target I_{D1}: $90\mu A$
- Calculating W/L

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{ov}^2$$

$$90\mu A = \frac{1}{2}(100\mu A/V^2)\left(\frac{W}{L}\right)(0.2V)^2$$

$$\frac{W}{L} = \frac{90}{50 \cdot 0.04} = 45$$

- Calculate g_m:

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2(90\mu A)}{0.2V} = 900\mu A/V = 0.9mS.$$

- Determining W & L

Tech spec: $L \geq 0.28nm$

Design spec: $0.5\mu m \leq L \leq 5\mu m$

Lets choose $L = 1\mu m$. Choosing one too little increases CLM (λ)
Choosing max ($5\mu m$) introduces parasitic capacitance which kill 500kHz bandwidth.

$$\therefore 45 = \frac{W}{L} \quad \begin{cases} W = 45\mu m \\ L = 1\mu m \end{cases}$$

- Calculating Drain resistor

└ L = 1μm

- Calculating Drain resistor
  └ Lets choose a value that results in a high stage gain $\leq 40dB$

  $A_{V_1} \approx -g_m \cdot R_D$

  └ Lets say $\sim 20 V/V$ gain (26dB)

  $20 = (0.9mS) \cdot R_D$

  $R_D \approx 22.2k$

- Biasing the Gate (V_G)

  $V_{GS} = V_{th} + V_{ov} = 0.63 + 0.2 = 0.83V$

  └ Therefore R_1 & R_2 must create 0.83V from provided 3.3V.

- Calculating R_1, R_2
  └ We now have two equations

  $V_G = V_{DD} \cdot \left(\frac{R_2}{R_1 + R_2}\right) = 0.83V$

  $R_{in} = R_1 \| R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} \geq 100k\Omega$ → for safety, assume $R_{in} = 150k\Omega$

  By the voltage ratio $\frac{0.83}{3.33} = 0.25$, we know $R_2 \approx \frac{1}{4} R_{Total}$

  Assume $R_1 = 600k\Omega$
  $R_2 = 200k\Omega$

  Check:

  $3.3 \cdot \frac{200}{800} = 0.825V$ (Close enough to 0.83V)

  $\frac{(600 \cdot 200)}{(600 + 200)} = 150k\Omega$ (Satisfies $\geq 100k\Omega$)

Stage 2 Design

- Lets take this to be common source again but this time coupling.
  └ The output of stage 1 is high DC; cannot be directly connected to gate of Stage 2.

- To reach a total gain of 1000 V/V (60dB), and assuming final output buffer has gain 0.9 V/V, stage 2:

  $A_{V_2} = \frac{1000}{20 \cdot 0.9} = 55 V/V$

- Assume NMOS again
  └ Use same I_D and V_{ov} for same reasons

- Calculating W/L, W & L

  $\frac{W}{L} = 45$ (from previous step)

  $L = 2\mu m$ → setting it to 2μm gives lesser CLM (λ)

  $W = 2 \cdot 45 = 90\mu m$

$W = 2 \cdot 45 = 90\mu m$

- Calculating gain

$A_{v2} = -g_m (R_{D2} \| r_{o2})$

└ g_{m2} is also $900\mu A$

$r_{o2} = \frac{1}{\lambda \cdot I_D}$; $\lambda = 0.04$ (from guidelines) $\approx \frac{1}{0.04 \cdot 90\mu A} = 277k\Omega$

└ If we target a gain of 50V/V

$50 = (900\mu A/V) \times R_{TOTAL}$

$R_{Total} = 55.5k\Omega$

$R_{total} = R_{D2} \| r_{o2}$; $R_{D2} = 70k\Omega$

Stage 3 Design

- We will use a common-Drain stage which will act as a buffer.
  - This is a solution to the load sensitivity requirement (gain reduction $\leq 10\%$) and the output swing requirement ($\geq 1.5\ V_{pp}$)
- Low output impedance ($R_{out} \approx 1/g_m$) to drive $10k\Omega$ load with minimal loss
- Use remainder of current budget.
  - └ Say $120\mu A$ for this stage

    $90 + 90 + 120 = 300\mu A$

    $300\mu A \cdot 3.3V = 0.99mW \leq 1mW$ (satisfied!)
- Use $V_{ov} = 0.2V$ again. to maximize g_m
- Dimensions W/L:

$$\frac{W}{L} = \frac{2 \cdot I_D}{\mu_n C_{ox} \cdot V_{ov}^2} = \frac{2 \cdot 120}{100 \cdot (0.2)^2} = 60$$

$L = 0.5\mu m$ for bandwidth, $W = 30\mu m$

- Transconductance:

$$g_m = \frac{2 I_D}{V_{ov}} = \frac{240\mu A}{0.2V} = 1.2mS$$

- Rout:

$$R_{out} \approx \frac{1}{g_{m3}} = \frac{1}{1.2mS} = 833\Omega$$

- Gain with load:

$$A_{v3} = \frac{R_L}{R_L + (1/g_{m3})} = \frac{10{,}000}{10000 + 833} = 0.923V/V$$

$\therefore$ There is a gain loss of $7.7\% \leq 10\%$ (satisfied!)